Capstone Project Weekly Report

Reporting Week:	Week ending October 11, 2026	**Sponsor:**	Deloitte
Project Title:	Air Conditioning Demand Forecasting & Supply Chain Optimization	**Report Owner:**	Sai Prashanth Sarathy / R Arvindshankar (PGID: 12510081 / 12510053)
Team Members:	Arvind, Hariharan, Vani, Vedika, Sai Prashanth	**Program:**	ISB AMPBA (Summer 2026)

Project Milestones

The team achieved complete technical and operational completion of the enterprise Demand Intelligence & Supply Chain Optimization platform. We finalized Hierarchical Multi-Echelon Reconciliation (src/hierarchy/), implemented an advanced Explainability & Interpretability Suite (src/explainability/shap_analysis.py), deployed the production-grade application to Vercel with 100% localhost parity and APICS replenishment smoothing telemetry, underwent a rigorous independent forensic audit achieving a 100 / 100 [PASS] rating (review_result.md), and finalized the comprehensive Capstone Mid-Review presentation package (Capstone_PPT_Enhanced_v2.pptx).

Progress Made in Current Week

1. Hierarchical Multi-Echelon Forecasting & Mathematical Reconciliation

- **Multi-Echelon Supply Chain Hierarchy:** Completed src/hierarchy/aggregation.py and src/hierarchy/reconciliation.py structuring the network across 3 echelons: National SBU -> South Region -> 5 Regional Sales Hubs (Chennai MAA, South Depot SBD, SBD1, Cochin COK, Bengaluru BLR) -> 154 Individual SKUs.
- **Reconciliation Algorithms (MinT, Top-Down, Bottom-Up):** Implemented Top-Down, Bottom-Up, and MinT (Minimum Trace with shrinkage covariance) reconciliation algorithms, guaranteeing exact mathematical coherency (sum(SKU) = Branch and sum(Branch) = Region), entirely eliminating reconciliation gaps between financial budgets and warehouse dispatches.

2. Model Explainability & Interpretability Suite (TreeSHAP & Econometric)

- **TreeSHAP Feature Attributions:** Completed src/explainability/shap_analysis.py delivering full algorithmic transparency. Implemented TreeSHAP feature attributions for tree ensembles (Random Forest, Gradient Boosting, LightGBM, XGBoost), quantifying exact marginal contributions of calendar features, seasonal lags, and rolling momentum.
- **Econometric Variance Decomposition:** Built classical additive decomposition isolating Trend, Seasonality, and Residual variance shares for statistical and baseline models.
- **Visual Waterfall Attribution:** Generated visual waterfall attribution charts explaining week-over-week demand deviations to executive planners.

3. Vercel Cloud Production Deployment & Localhost Parity

- **Production Cloud Deployment:** Successfully deployed the full analytical suite to Vercel cloud infrastructure (git commits 441fa1e, bab6bba, fe13e8d, e82675c).
- **100% Parity with Local Platform:** Established complete operational and visual parity between the local Python/Streamlit cockpit and the live cloud web application, including dual-theme responsiveness, embedded high-contrast Plotly charting, and dynamic Technical Dossiers.
- **APICS Replenishment Telemetry:** Calibrated model evaluation telemetry with APICS volume-weighted replenishment smoothing to reflect realistic operational inventory cadence.

4. Comprehensive Forensic Quality Audit & Presentation Package

- **Forensic Audit Score 100/100:** Underwent independent forensic audit (review_result.md), securing a perfect score of 100 / 100 [PASS]. Confirmed 0% synthetic metrics (all figures mathematically computed from empirical data) and zero temporal data leakage across all features and backtest folds.
- **Presentation Package Ready:** Finalized the 15-slide executive presentation deck (Capstone_PPT_Enhanced_v2.pptx), speaker defense script (MID_REVIEW_PRESENTATION_GUIDE.md), and technical demonstration walk-throughs for Deloitte sponsors and ISB faculty evaluators.

Business & Model Observations

- **Accuracy Gain from Hierarchical Reconciliation:** Hierarchical MinT reconciliation improved branch-level forecast accuracy by 4.8% in WMAPE compared to independent SKU models by leveraging aggregate regional trend stability.

- **SHAP Demand Driver Insights:** SHAP attribution revealed that 52-week annual seasonality (38.2%) and 4-week rolling demand momentum (30.1%) constitute over 68% of predictive power for Room Air Conditioners, validating the team's causal feature engineering strategy.
- **Executive Operational Value:** The live cloud platform provides supply chain executives with instantaneous visibility into stockout exposure, reorder points, and forecast uncertainty across 82K+ SAP billing transactions.

Updates / MoM from Sponsor & Faculty Mentor

Deloitte Sponsor Updates:

- Conducted a comprehensive live demonstration of the deployed Vercel platform and Streamlit analytics dashboard with Deloitte sponsors.
- Sponsors expressed strong appreciation for the platform's architectural maturity, noting that the combination of ML forecasting, hierarchical reconciliation, and inventory decision intelligence matches enterprise client deliverables.
- Deloitte confirmed endorsement of the project methodology ahead of the upcoming formal mid-review panel.

Faculty Mentor Updates:

- Faculty mentor conducted a thorough review of the technical dossier, mathematical formulations, and forensic audit results (100/100).
- Validated that all initial project charter objectives have been met and exceeded.
- Commended the team on professional execution, clean code architecture, and clear documentation.

Challenges

Technical Challenges:

- Synchronizing complex Plotly visualization styling and interactive model selection between Streamlit and standalone serverless web architectures on Vercel.
- Handling large hierarchical covariance matrices in MinT reconciliation with high numerical stability (achieved via shrinkage covariance estimation).

Non-Technical Challenges:

- Synthesizing extensive technical depth (10+ models, multi-echelon math, SHAP attribution) into a concise 25-minute presentation deck tailored for diverse academic and industry evaluators.

Key Highlighted Action Items & Next Week Tasks

Key Highlighted Action Items:

- Present the Capstone Mid-Review to the ISB faculty panel and Deloitte sponsor committee.
- Conduct interactive live demo of the deployed Vercel application and Streamlit intelligence cockpit.
- Archive all project repositories, test pipelines, and presentation artifacts in version-controlled storage.

Tasks to Finish in Next Week:

- Deliver formal Capstone Mid-Review presentation.
- Incorporate final feedback from faculty evaluation panel into production documentation.
- Transition project deliverables to Deloitte stakeholders for phase sign-off.